Algorithms for Robust Dedicated Actuator Placement

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Abstract

We study dedicated actuators installed at distinct states of a structured linear system. Failures remove actuators but preserve states and state arcs. For unit costs and one failure, an exact source-quota relaxation followed by maximum matching in a *common-repair graph* yields a 3/2-approximation on arbitrary directed state graphs using $O(n^3)$ elementary graph operations. The analysis combines fundamental cocircuit unions with the Tutte–Berge formula. For multiple failures, we give an exact algorithm fixed-parameter tractable in the maximum-matching deficiency and failure budget, independently of the number of source strongly connected components. The underlying result solves resilient matroid spanning with an arbitrary number of disjoint unit-cost quota constraints. We also establish an exact weighted spanning algorithm for safe and vulnerable elements in the independence-oracle model, with an application to strongly connected state graphs.

Keywords: structural controllability; dedicated actuators; fault tolerance; matroid spanning; approximation algorithms; fixed-parameter tractability.

1 Introduction

A structurally controllable system can lose controllability when an actuator fails. Redundant placement must protect both the matching condition and accessibility from the installed inputs. These requirements have different combinatorial forms: robust spanning in a matroid and lower quotas on source strongly connected components. Keeping both requirements explicit permits approximation and exact parameterized algorithms within the same model.

First, we give a cubic-time 3/2-approximation for one actuator failure and unit costs (Theorem 4.5). The repair argument applies to arbitrary matroids, while an exact graph relaxation supplies the required lower bound. Example 4.6 attains the ratio for one optimal solution of the relaxation.

Second, we give an exact fixed-parameter algorithm for weighted resilient spanning in arbitrary independence-oracle matroids, allowing both safe and vulnerable elements (Theorem 5.2). Extending the uniform-set core method of Bentert et al. [5], a weight-ordered recursion preserves an optimum selected set, including after a large uniform set has been found. A set-pairs inequality sharpens the hyperplane count. The parameter is matroid rank plus failure budget, and the objective charges every selected element.

Third, we solve exact unit-cost resilient spanning with arbitrarily many disjoint quota groups (Theorem 6.2). Because a bounded resilient core may omit elements needed for accessibility, the construction also preserves quota-completion cost through groupwise representative caps. This yields an algorithm fixed-parameter tractable in matching deficiency and failure budget on arbitrary digraphs (Corollary 6.3), without parameterizing the number of source components.

Poças, Pequito, and Ramos [14] establish NP-hardness and approximation guarantees of 5/2 on general digraphs and 2 on strongly connected digraphs for the robust minimal structural controllability problem. These are reference points rather than like-for-like comparisons without specifying actuator and failure models. Section 2 distinguishes the relevant earlier work. Section 3 fixes the model and matroid reduction; Section 4 gives the approximation algorithm; Sections 5 and 6 develop the two exact parameterized algorithms; and Section 7 concludes.

2 Related work and model distinctions

The robust purchased-set objective differs from choosing an ordinary basis after failures. Bentert et al. [5] prove unweighted FPT in rank plus failure budget and explicitly pose the weighted question. In the ESA 2025 proceedings version of [5], Proposition 6 gives the layered cardinality bound, Lemma 8 the uniform-set exchange and basis-subset hyperplane count, and Lemma 9 the recursive core. These are inherited ingredients. Our weighted argument adds the ordered exchange invariant, recursion in both stopping cases, and separation of failure statuses. The sharper count is an application of Bollobás's classical set-pairs inequality [6]. The same paper also gives polynomial weighted algorithms for rank at most two and for truncations of unit-capacity partition matroids.

The all-vulnerable case of Theorem 5.2 addresses the weighted selected-set problem posed in Section 1.1 of [5]. Section 7 of its ESA proceedings version prints the condition $\operatorname{rank}(M-F) = \operatorname{rank}(M)$ in a separate formulation; that condition should be distinguished from $\operatorname{rank}_M(B \setminus F) = \operatorname{rank}(M)$, the selected-set condition used here and in related fault-tolerant design [3]. We make no claim about the intended interpretation of the printed text.

Theorem 7 of Adjiashvili, Bindewald, and Michaels [2] gives a 7/4-approximation for uniform-cardinality node-robust assignment. Their repair perspective anticipates the approach here. Theorem 5 of their ICALP 2016 paper [1] gives 3/2 for uniform *edge*-robust assignment, buying assignment edges and tolerating an edge failure via ear decompositions. That is a different purchase/failure model. The proof in Section 4 instead uses fundamental cocircuit unions, maximum common-repair matching, and source quotas.

Bulk-robust optimization [4] charges purchased elements but supplies an explicit family of failure scenarios. Expanding all at most k failures can require $n^{\Theta(k)}$ scenarios, so polynomial time in that expanded list does not establish the present FPT bound. Weighted representative families [8] preserve independent extensions under their stated representation and matroid hypotheses; their predicate is not directly optimum preservation for a dependent purchased set under all failures. Likewise, the cheapest-surviving-basis preservers of Dey and Kavitha [7] optimize a different object, as Example 5.10 shows.

3 Model and the dual matching matroid

Let $G = (E, \mathcal{A})$ be a nonempty finite directed state graph with $n = |E|$; self-loops are allowed. Each prescribed nonzero entry of the state and input matrices is a separate free parameter. Structural controllability means that a controllable numerical realization exists, equivalently that almost every realization is controllable. A placement $P \subseteq E$ installs one dedicated input at each selected state, at most one per state. An actuator failure removes the installed input while the state vertices and arcs remain intact; the placement is fixed before failures occur.

Write $S_1, \ldots, S_s$ for the source strongly connected components (SCCs), those having no entering state arc from another SCC. Form the state bipartite graph with a left and right copy of E, and an edge $u_L v_R$ for each arc $u \to v$. Let $\mathcal{T}$ be the transversal matroid on the right copies: its independent

sets are the right-state sets matchable by bipartite edges. Put $M = \mathcal{T}^*$ and $d = n - \mathrm{rank}_{\mathcal{T}}(E)$, the maximum-matching deficiency. We identify each state, its right copy, and the corresponding element of M when no ambiguity can arise. We write rank_M when the matroid must be specified, and rank when it is clear from context. In general matroid statements the rank is r; for this control construction, $r = d$. For $X \subseteq E$, duality gives

$$\mathrm{rank}_M(X) = |X| + \mathrm{rank}_{\mathcal{T}}(E \setminus X) - \mathrm{rank}_{\mathcal{T}}(E). \tag{1}$$

In particular, X spans M if and only if $E \setminus X$ is independent in $\mathcal{T}$. A rank oracle for M is computable by bipartite matching.

We call a placement *k-robust* if every failure of at most k selected actuators leaves the system structurally controllable. For matroid spanning we use *resilient* for the analogous property, with the failing elements specified explicitly. Throughout, an *actuator* is a dedicated input assigned to one state.

The standard structural-controllability criterion combines accessibility with a matching saturating all state vertices [10, 13]. For dedicated actuators it gives the following equivalence.

Proposition 3.1. *A placement P is k-robust if and only if*

$$\mathrm{rank}_M(P \setminus F) = d \quad (F \subseteq P,\ |F| \le k), \qquad |P \cap S_i| \ge k+1 \quad (1 \le i \le s). \tag{2}$$

Proof. For a fixed surviving placement X, matching every state in $E \setminus X$ by state edges and every state in X by its own input gives a matching saturating all states. Conversely, an input-augmented saturating matching matches every state in $E \setminus X$ using state edges. This proves the matching equivalence, and (1) gives spanning in M. Accessibility holds exactly when every source SCC contains an input: every component is reachable from a source component in the condensation DAG, and no input outside a source component can reach it. The source SCCs retain an input after any failure of at most k actuators exactly when they initially contain at least $k+1$ inputs. Applying both criteria to every survivor proves the claim. □

We use standard finite-matroid notation. The closure of X is $\mathrm{cl}(X) = \{e : \mathrm{rank}(X \cup \{e\}) = \mathrm{rank}(X)\}$. A hyperplane is a flat of rank one below the matroid rank. A loop has rank zero; a coloop belongs to every basis. The coloops of a restriction $M|P$ are precisely the elements whose deletion lowers $\mathrm{rank}_M(P)$. A circuit is a minimal dependent set. For a basis B and $e \notin B$, the unique circuit contained in $B \cup \{e\}$ is its fundamental circuit. The *fundamental cocircuit* at $b \in B$ is $E \setminus \mathrm{cl}(B \setminus \{b\})$ [12]. We use $U_{r,m}$ for the rank-r uniform matroid on m elements; a partition matroid limits how many elements may be chosen from each disjoint class, and its rank-r truncation replaces every independent set of size greater than r by a dependent set. A parallel class consists of nonloop elements with the same rank-one closure.

Problem 3.2 (Robust dedicated-actuator placement)**.** Given a state digraph, an actuator-failure budget k, and available distinct-state locations, find a minimum-cost k-robust placement or certify infeasibility. The unit-cost case minimizes the number of installed actuators.

4 A cubic-time 3/2-approximation

Throughout this section $k = 1$ and every actuator has unit cost.

4.1 An exact source-quota relaxation

Minimize $|P|$ subject to P spanning M and $|P \cap S_i| \geq 2$ for all i. Denote its optimum by L. Every robust placement is feasible for this relaxation, so $L \leq \mathrm{OPT}$.

The relaxation reduces to an integral maximum-flow problem. The complement $I = E \setminus P$ must be matchable using state edges and must satisfy $|I \cap S_i| \leq |S_i| - 2$. The network has a super-source, two state copies, one group node per source SCC, and a super-sink. Edges from the super-source to left-state vertices and bipartite state edges have unit capacity. Each right-state vertex has a unit-capacity outgoing edge to its group node if it belongs to a source SCC, and to the super-sink otherwise. The edge from group node i to the super-sink has capacity $|S_i| - 2$. An integral maximum flow gives a maximum-cardinality admissible I, and therefore a minimum-cardinality P.

Proposition 4.1. *In the one-failure model, a unit-cost robust placement exists if and only if every source SCC contains at least two states. In this case the dual matching matroid M has no coloops.*

Proof. A singleton source SCC cannot retain an actuator after one failure. Conversely, if all source SCCs have at least two vertices, every state has an incoming state arc: this holds inside each nonsingleton SCC, while a singleton nonsource SCC has an entering arc. Thus $\mathcal{T}$ has no loops and its dual M has no coloops. The full placement E spans after any single deletion and satisfies all source quotas. □

4.2 Cocircuit unions and a global lower bound

The next two lemmas concern an arbitrary rank-d matroid M with a spanning set P. Let D be the coloops of $M|P$. For $b \in D$ define

$$C_b = E \setminus \mathrm{cl}(P \setminus \{b\}).$$

The set C_b is the fundamental cocircuit at b with respect to any basis of P. It meets P exactly in $\{b\}$, and an external element of C_b restores spanning after deleting b.

Lemma 4.2. *For every $R \subseteq D$,*

$$\mathrm{rank}\Big(E \setminus \bigcup_{b \in R} C_b\Big) = d - |R|.$$

Proof. Choose a basis $B \subseteq P$. Every $b \in D$ belongs to B. Since $B \setminus \{b\} \subseteq P \setminus \{b\}$ and both sets have rank $d - 1$, their closures coincide. For nonempty R we claim

$$\bigcap_{b \in R} \mathrm{cl}(B \setminus \{b\}) = \mathrm{cl}(B \setminus R).$$

For $e \notin B$, membership in $\mathrm{cl}(B \setminus \{b\})$ means that its fundamental circuit omits b. Membership for every $b \in R$ therefore means that this circuit is contained in $(B \setminus R) \cup \{e\}$, which is precisely membership in $\mathrm{cl}(B \setminus R)$. Basis elements satisfy the equality directly, and taking ranks proves the lemma. For $R = \varnothing$, the statement reduces to $\mathrm{rank}(E) = d$. □

Form the *common-repair graph* K, a simple undirected (possibly nonbipartite) graph on D, with an edge bc when some $e \in E \setminus P$ belongs to both C_b and C_c. Let ν denote its maximum matching cardinality.

Lemma 4.3. *Every set W spanning M after any one deletion satisfies $|W| \geq 2(|D| - \nu)$.*

Proof. Fix $U \subseteq D$ and let $R_1, \ldots, R_t$ be the nonempty connected components of $K - U$. Put $Q_i = \bigcup_{b \in R_i} C_b$. These sets are pairwise disjoint. An element of P belongs to at most one C_b. A common external element of Q_i, Q_j would witness an edge between their components. Lemma 4.2 gives $\operatorname{rank}(E \setminus Q_i) = d - |R_i|$. Since adding an element raises rank by at most one, W must contain at least $|R_i|$ elements of Q_i to span. Equality is impossible: deleting one of these elements would leave rank at most $(d - |R_i|) + (|R_i| - 1) < d$. Thus

$$|W| \geq \sum_i (|R_i| + 1) = |D| - |U| + t \geq |D| - |U| + \operatorname{odd}(K - U),$$

where odd counts odd-order components. The classical Tutte–Berge formula [11] states

$$2\nu = |D| - \max_{U \subseteq D} \big(\operatorname{odd}(K - U) - |U|\big).$$

Taking the maximum in the preceding inequality proves the result. □

4.3 Algorithm and guarantee

Algorithm 4.4 (Common-repair augmentation)**.** 1. Reject if a singleton source SCC exists. Otherwise, solve the source-quota flow relaxation for an optimal set P.

2. Find the coloops D of $M|P$, construct K, and retain one external common-repair witness for each edge of K.

3. Find a maximum matching in K. Select the stored witness for each matched pair and any external element of C_b for each unmatched $b \in D$. Return P together with these witnesses, removing duplicates.

Theorem 4.5. *For any nonempty directed state graph in the stated dedicated-actuator model, a robust placement of size at most $\frac{3}{2}\mathrm{OPT}$ can be found, or infeasibility certified, in $O(n^3)$ elementary graph operations and $O(n^2)$ space. The guarantee is for unit costs and one actuator failure.*

Proof. By Proposition 4.1, Algorithm 4.4 rejects exactly the infeasible instances. For a feasible instance, let P be an optimal solution of the flow relaxation and let R_{rep} be the selected witness set, with duplicates removed. Every unmatched $b \in D$ has an external witness because M has no coloops while $P \setminus \{b\}$ has rank $d - 1$. At most $\nu + (|D| - 2\nu) = |D| - \nu$ witnesses are selected.

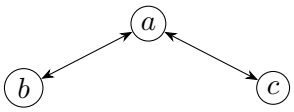
The set $P \cup R_{\mathrm{rep}}$ is robust. If a witness fails, P still spans. If an original noncritical element fails, the surviving part of P spans. If $b \in D$ fails, its witness repairs the deficiency. The source quotas persist under additions. By Lemma 4.3,

$$|P \cup R| \leq L + |D| - \nu \leq \mathrm{OPT} + \frac{\mathrm{OPT}}{2}.$$

For the complexity, retain the state matching J obtained from the quota flow. It covers $I = E \setminus P$. For each $b \in P$, explore alternating paths from unmatched right vertex b: traverse state edges to the left, then matched edges back to the right. An alternating path reaching a free left vertex is an augmenting path witnessing that $I \cup \{b\}$ is matchable, so b is noncritical. For a critical b, the reachable matched right vertices are exactly its external repairs. A path to right vertex e can be flipped to match $(I \setminus \{e\}) \cup \{b\}$. Conversely, the symmetric difference with any matching of that replacement set gives such a path, since a path to a free left vertex would contradict criticality. Hence these searches recover the complete repair supports.

There are $O(n)$ vertices and $O(n^2)$ edges in the flow network. At most n unit augmentations, each found in $O(n^2)$ operations, suffice. At most n alternating searches cost $O(n^3)$ in total. For each external state, connect all pairs of critical states it repairs, storing a witness for each edge. This clique construction costs $O(n^3)$. A maximum general-graph matching is computable in $O(n^3)$ operations by a blossom algorithm [9, 11]; a maximal matching is insufficient. All structures require $O(n^2)$ space. □

Example 4.6 (Three-state bidirected star)**.** Consider the bidirected star below, with center a and leaves b, c.



The relaxation may choose $P = \{a, c\}$, whose only critical state is c and whose only external repair is b. Algorithm 4.4 returns all three states, whereas $\{b, c\}$ is robust. Thus the factor $3/2$ is attained for this choice of optimal relaxation solution. Selecting the robust pair directly in the relaxation avoids this case.

5 Weighted resilient matroid spanning

Let $M = (E, \mathcal{I})$ be a rank-r independence-oracle matroid, $V \subseteq E$ the vulnerable elements, $E \setminus V$ the safe elements, and $k \geq 0$ the failure budget. A set B is *resilient* if

$$\operatorname{rank}_M(B \setminus F) = r \quad \text{for every } F \subseteq B \cap V, \quad |F| \leq k. \tag{3}$$

For nonnegative binary-encoded weights w, the objective is to minimize $w(B) = \sum_{e \in B} w(e)$. Here resilience concerns the selected set B, including when some elements of E remain unselected.

Problem 5.1 (Weighted resilient spanning)**.** Given an independence-oracle matroid M, weights $w \geq 0$, a vulnerable subset V, and budget k, find a minimum-weight resilient set or certify infeasibility.

Theorem 5.2. *An optimum resilient set can be computed, or infeasibility certified, in $f(r,k)\,\mathrm{poly}(n, \ell)$ time and independence-oracle calls, where $n = |E|$, ℓ is the input bit length, and each independence query has unit cost. For $r \geq 2$, writing $q = (k+1)r$, one may take*

$$f(r,k) = 2^{O(qr^2(r+k))},$$

hence the running time is at most $2^{O((r+k)^5)}\,\mathrm{poly}(n, \ell)$. The cases $r \leq 1$ or $k = 0$ are polynomial-time solvable.

5.1 Matroid preliminaries and a small optimum

Loops can be discarded because their removal preserves resilience and cannot increase nonnegative purchase cost. If $r = 0$, the empty set is optimal. If $k = 0$, a minimum-weight basis is optimal by the matroid greedy algorithm. For $r = 1$, choose the cheaper feasible option between a cheapest safe nonloop and the $k + 1$ cheapest vulnerable nonloops. Henceforth assume $r \geq 2$ and $k \geq 1$, and work on the loop-free restriction. Set

$$q = (k+1)r, \qquad H = \binom{r+k-1}{k}, \qquad t_h = (h-1)H + q \quad (1 \leq h \leq r). \tag{4}$$

Lemma 5.3. *Every resilient set B contains a resilient subset B_0 of cardinality at most q. Consequently, if a feasible solution exists, there is a minimum-weight solution of cardinality at most q.*

Proof. Successively choose a maximal independent subset I_j of $B \setminus \bigcup_{i<j} I_i$, for $j = 0, \dots, k$, allowing empty layers, and put $B_0 = \bigcup_{j=0}^{k} I_j$. Since $|I_j| \le r$, we have $|B_0| \le q$.

Fix $F \subseteq B_0 \cap V$ with $|F| \le k$. The layers are disjoint, so some I_j avoids F. Every element of $B \setminus B_0$ was available when I_j was chosen and belongs to $\mathrm{cl}(I_j)$. Every element of $B_0 \setminus F$ already lies in that surviving set. Therefore

$$B \setminus F \subseteq \mathrm{cl}(B_0 \setminus F).$$

Since F is also an allowed failure of B, its left-hand side spans M. Thus B_0 is resilient. Nonnegative weights give $w(B_0) \le w(B)$, and applying this construction to an optimum proves the second assertion. □

5.2 A weight-respecting exchange

The set-pairs inequality of Bollobás [6] yields the following sharp bound for deficient hyperplanes.

Lemma 5.4. *For any $Y \subseteq E$, at most $\binom{r+k-1}{k}$ distinct hyperplanes can have the form $\mathrm{cl}(Y \setminus F)$ with $F \subseteq Y$ and $|F| \le k$. This count is sharp.*

Proof. Write $\Pi_1, \dots, \Pi_m$ for these distinct hyperplanes. Choose a basis $A_i \subseteq Y \cap \Pi_i$ of size $r-1$, and put $\Delta_i = Y \setminus \Pi_i$. Because $\Pi_i = \mathrm{cl}(Y \setminus F)$ for some permitted F, we have $\Delta_i \subseteq F$, so $|\Delta_i| \le k$. Also $A_i \cap \Delta_i = \varnothing$. If $i \ne j$, then $A_i \cap \Delta_j \ne \varnothing$: otherwise $A_i \subseteq \Pi_j$ and hence $\Pi_i = \mathrm{cl}(A_i) \subseteq \Pi_j$, forcing equality of the two hyperplanes.

In a uniformly random permutation of Y, let $\mathcal{E}_i$ be the event that every element of A_i precedes every element of Δ_i. Its probability is $\binom{r-1+|\Delta_i|}{r-1}^{-1}$. These events are pairwise disjoint. Indeed, for $i \ne j$ choose $a \in A_i \cap \Delta_j$ and $b \in A_j \cap \Delta_i$. The two events require a before b and b before a, respectively. Therefore

$$\sum_i \binom{r-1+|\Delta_i|}{r-1}^{-1} \le 1.$$

Each summand is at least $\binom{r+k-1}{r-1}^{-1}$, proving the bound. For sharpness take $M = U_{r,r+k}$, $B = E$, all elements vulnerable, and $Y = B \setminus \{x\}$. The set B is resilient, and each k-subset of Y leaves a distinct $(r-1)$-element hyperplane. This gives exactly $\binom{r+k-1}{k}$ distinct hyperplanes of the required form. □

Call Z *h-uniform* if $\mathrm{rank}(Z) = h$ and every h-element subset of Z is independent. Such a set has at least h elements, and every subset of size at most h is independent. A set has a single failure status if it lies entirely in V or entirely in $E \setminus V$.

Lemma 5.5. *Let $A \subseteq E$ have a single failure status and rank $h \ge 1$. Let $Z \subseteq A$ be h-uniform with $|Z| \ge t_h$. Suppose B is resilient, $|B| \le q$, and $x \in (B \cap A) \setminus Z$ satisfies $w(z) \le w(x)$ for every $z \in Z$. Then some $y \in Z \setminus B$ makes*

$$B' = (B \setminus \{x\}) \cup \{y\}$$

resilient, with $|B'| = |B|$ and $w(B') \le w(B)$.

Proof. Put $Y = B \setminus \{x\}$. Consider the permitted failures $F \subseteq Y \cap V$, $|F| \le k$, for which $\mathrm{rank}(Y \setminus F) < r$. Resilience of B implies that their surviving ranks equal $r-1$ and that $x \notin \mathrm{cl}(Y \setminus F)$. Lemma 5.4 bounds the number of these distinct hyperplanes by H. It applies also when x is safe and some deficient failure sets have size less than k.

Each of these hyperplanes contains at most $h-1$ elements of Z. Otherwise it contains an independent h-subset of Z, which spans A, and hence contains x, a contradiction. At most $q-1$ elements of Z belong to B, because $x \notin Z$. The number of forbidden choices in Z is at most $(h-1)H + q - 1 < t_h$. Choose a $y \in Z \setminus B$ outside all the deficient hyperplanes.

Let $F' \subseteq B' \cap V$ have size at most k. If $y \notin F'$, then $F' \subseteq Y$; either $Y \setminus F'$ already spans or y repairs its rank deficiency by construction. If $y \in F'$, then y is vulnerable, so x is vulnerable as well. The set

$$(F' \setminus \{y\}) \cup \{x\}$$

is an allowed failure of B of the same size as F'. Its survivor is exactly $B' \setminus F'$, which therefore spans. The cardinality and weight assertions follow from the choice of y. □

5.3 Recursive construction of an optimum-preserving core

For a loop-free set A of a single failure status, construct $\mathrm{Core}(A)$ by the following procedure. Every scan uses a fixed total ordering refining nondecreasing weight.

Algorithm 5.6 (Weight-ordered core $\mathrm{Core}(A)$)**.** 1. If $A = \varnothing$, return $\varnothing$. Set $h = \mathrm{rank}(A)$. If $|A| \leq t_h$, return A.

2. Scan the elements of A in order, initially with $Z = \varnothing$. Accept an element x precisely when every subset of $Z \cup \{x\}$ of size at most h is independent. Stop when $|Z| = t_h$ or all elements have been scanned.

3. If $h = 1$, return Z. Otherwise, for every $J \in \binom{Z}{h-1}$, put $A_J = A \cap \mathrm{cl}(J)$ and recursively compute $\mathrm{Core}(A_J)$. Return

$$Z \cup \bigcup_{J \in \binom{Z}{h-1}} \mathrm{Core}(A_J).$$

The recursive step applies whether the scan reaches its threshold or exhausts A. The parameters q, H remain fixed across recursive calls.

Until Z has size h, the acceptance rule is ordinary greedy independence. The final Z therefore has at least h elements and is h-uniform. Each child has rank exactly $h-1$: it contains T and is contained in $\mathrm{cl}(T)$.

Lemma 5.7. *For every resilient B with $|B| \leq q$ and every single-status loop-free set A, there is a resilient B' such that*

$$|B'| = |B|, \quad w(B') \leq w(B), \quad B' \setminus A = B \setminus A, \quad B' \cap A \subseteq \mathrm{Core}(A).$$

Proof. Induct on $h = \mathrm{rank}(A)$. Empty domains and a return of all of A require no change. For $h = 1$, every scanned nonloop is accepted, so the threshold scan retains the q lightest elements. Lemma 5.5 replaces every selected element outside Z at no greater weight, proving the base case.

Suppose $h > 1$ and the scan is executed. Let $U = \bigcup_J A_J$. Every rejected scanned element x belongs to U. Indeed, at rejection a dependent subset of $Z \cup \{x\}$ contains x and at most $h-1$ retained elements. Those retained elements are independent, so they span x. Extend them to an $(h-1)$-subset of the final Z, obtaining a child containing x. Also $Z \subseteq U$, since $h > 1$ and every element of Z belongs to an $(h-1)$-subset of Z.

If the scan is exhausted, $A = U$. Otherwise each element of $A \setminus U$ is unscanned and is at least as expensive as every element of Z. Apply Lemma 5.5 successively to the elements of $B \cap (A \setminus U)$. Each replacement lies in $Z \subseteq U$ and removes one selected element outside U. The resulting resilient set has its intersection with A inside U, unchanged cardinality and outside-A part, and no larger weight.

Fix the final output $W_A = \mathrm{Core}(A)$ and process the children $A_1, \dots, A_m$ in any order. Apply the induction hypothesis to each child and the current solution. For the current solution define $\Psi(B) = (B \cap A) \setminus W_A$. Processing A_i changes only elements in A_i and leaves its selected elements in $\mathrm{Core}(A_i) \subseteq W_A$. Consequently,

$$\Psi(B_{\mathrm{new}}) \subseteq \Psi(B_{\mathrm{old}}) \setminus A_i.$$

Initially $\Psi(B) \subseteq U$. After all children have been processed it is empty, regardless of overlaps between children. All operations preserve cardinality, hence the bound q, and preserve resilience without increasing weight. This establishes the induction. □

Apply Algorithm 5.6 separately to the vulnerable and safe nonloop elements, and let W be the union of its outputs.

Proposition 5.8. *Whenever the weighted resilient-spanning instance is feasible, W contains an optimum selected set of cardinality at most q.*

Proof. The two core domains are disjoint. Lemma 5.3 gives an optimum of size at most q, and applying Lemma 5.7 to each domain preserves resilience and total weight while moving every selected element into the corresponding output. □

5.4 Size and running time

Put $N_0 = 0$, $N_1 = q$, and, for $2 \le h \le r$, define

$$N_h = t_h + \binom{t_h}{h-1} N_{h-1}.$$

By induction, a rank-h call returns at most N_h elements. The sequence is nondecreasing; in particular $|W| \le 2N_r$, or $|W| \le N_r$ if all elements are vulnerable.

For an explicit bound put $\tau = t_r$ and $N = 2r\tau^{(r-1)^2}$. Each of the two recursion trees has at most r levels and branching at most τ^{r-1}, giving at most N nodes and $|W| \le N\tau$. Each node uses $O(n(1+\tau^{r-1}))$ independence queries for the uniformity scan and $O(n\tau^{r-1})$ queries for its lower-rank closures. Rank computation and sorting require polynomial input time at each node, with factors depending only on r and k.

The algorithm enumerates all $B \subseteq W$ with $|B| \le q$; for each candidate it checks all allowed failures and computes the surviving ranks using the independence oracle. There are at most $(N\tau+1)^q$ candidates and at most 2^q failures per candidate. Choose a feasible candidate of minimum total weight. If there is none, optimum preservation proves infeasibility. Exact weight sums and comparisons have polynomial bit cost in the original weight encoding.

For $r \ge 2$, put $p = r + k$. Since $H \le 2^{p-1}$ and $q \le (p+1)^2$, we have $\log_2 \tau = O(p)$. Hence $N\tau = 2^{O(r^2 p)}$, and enumerating the candidates and failures gives $2^{O(qr^2p)}\,\mathrm{poly}(n,\ell)$. As $q = O(p^2)$, this is at most $2^{O(p^5)}\,\mathrm{poly}(n,\ell)$. For fixed $k \ge 1$, the sharper estimate $H = O_k(r^k)$ gives $\tau = O_k(r^{k+1})$, $N\tau = r^{O_k(r^2)}$, and running time $r^{O_k(r^3)}\,\mathrm{poly}(n,\ell)$.

Proof of Theorem 5.2. The cases $r = 0$, $r = 1$, and $k = 0$ follow from the preliminary algorithms. For $r \ge 2$ and $k \ge 1$, Lemma 5.3 bounds the size of an optimum, Lemma 5.5 permits weight-respecting replacements, and Proposition 5.8 ensures that the output of Algorithm 5.6 preserves an optimum. The enumeration just described therefore finds a minimum-weight resilient set, or certifies infeasibility. The derived bound gives the claimed fixed-parameter running time and polynomial bit complexity. □

5.5 Why simpler weighted reductions do not suffice

Example 5.9 (Premature uniform-scan termination)**.** Consider a rank-two truncation of a partition matroid with two parallel classes of size two having weights $(0, 0)$ and $(1, 1)$, and five further singleton classes each of weight 100. Let $k = 1$. Then $q = 4$, $H = 2$, and $t_2 = 6$. The weight-ordered uniform scan reaches the threshold while retaining only one element from each cheap parallel class. The minimum resilient cost within the retained uniform set is 101, whereas the full instance has optimum 2, attained by the two cheap pairs. Thus sorting the scan is insufficient if a successful threshold causes the recursion to stop. Our construction retains the cheap rejected elements through the lower-rank children.

Example 5.10 (Ordinary-basis preservers)**.** Dey and Kavitha [7] study a preserver $S \subseteq E$ such that, for every $F \subseteq E$ with $|F| \leq k$, $M|(S \setminus F)$ contains a minimum-cost ordinary basis of $M|(E \setminus F)$. That property need not preserve the optimum total cost of a resilient purchased set.

Consider a five-element rank-two truncation with two parallel classes $\mathcal{P}, \mathcal{Q}$, each of size two, and singleton class $\mathcal{R}$. Every element of $\mathcal{P} \cup \mathcal{Q}$ has weight 2 and the element of $\mathcal{R}$ has weight 3. After any failure of at most one ground element, $\mathcal{P} \cup \mathcal{Q}$ still contains an ordinary basis of cost 4, which is minimum in the full surviving matroid. Yet the resilient optimum inside $\mathcal{P} \cup \mathcal{Q}$ is 8, and the full instance has optimum 7, using one element from each class. Two successive cheapest residual bases also produce $\mathcal{P} \cup \mathcal{Q}$. Thus this preserver construction does not replace the optimum-preserving recursion above.

5.6 An application to dedicated-actuator placement

Consider a nonempty strongly connected state digraph with nonnegative actuator costs and specified safe or vulnerable locations. Keep the single-actuator-per-state and actuator-only failure conventions of Section 3. Its dual state matching matroid has rank equal to the deficiency d; Proposition 3.1 supplies the matching interpretation. When only vulnerable actuators may fail, its rank condition is applied to $F \subseteq P \cap V$; the same surviving-input matching argument proves this restriction.

Corollary 5.11. *Under the stated failure model, minimum-cost robust dedicated-actuator placement on a strongly connected state digraph is fixed-parameter tractable in $d + k$, for nonnegative binary-encoded costs and arbitrary assignments of safe and vulnerable actuator locations.*

Proof. If $d > 0$, every surviving spanning set in M is nonempty. Strong connectivity then ensures accessibility, so Theorem 5.2 applies. If $d = 0$, matching is automatic and accessibility requires a nonempty survivor. Choose the cheaper of one cheapest safe actuator and $k + 1$ cheapest vulnerable actuators, omitting unavailable options. □

For general state digraphs, source-SCC constraints are also necessary. The weighted same-status exchange alone does not preserve them. Section 6 gives a unit-cost quota theorem; no weighted-quota extension is claimed.

6 Exact parameterized optimization with source quotas

On a general state digraph, spanning must be combined with source-SCC accessibility quotas. The following matroid theorem permits an arbitrary number of disjoint quota groups, with unit costs and all elements vulnerable.

Problem 6.1 (Quota-constrained resilient spanning)**.** Given a rank-r independence-oracle matroid, disjoint quota groups $S_1, \dots, S_s$, and budget k, find a minimum-cardinality set that spans after any failure of at most k selected elements and contains at least $k+1$ elements of each group, or certify infeasibility.

Theorem 6.2. *Let M be a rank-r matroid on E, given by an independence oracle, and let $S_1, \dots, S_s \subseteq E$ be pairwise disjoint. For $k \geq 0$, a minimum-cardinality set B satisfying*

$$\operatorname{rank}_M(B \setminus F) = r \quad (F \subseteq B,\ |F| \leq k), \qquad |B \cap S_i| \geq k+1 \quad (1 \leq i \leq s)$$

can be computed, or infeasibility certified, in $f(r,k)\,\mathrm{poly}(|E|)$ oracle and arithmetic operations.

Proof. Set $\beta = k+1$, $q = \beta r$, and $G_0 = E \setminus \bigcup_i S_i$. A group with fewer than β elements makes the instance infeasible. For $r = 0$, the optimum consists of β elements from each group. Assume $r \geq 1$.

By Lemma 5.3, a resilient spanning set has a resilient subset C of size at most q, which can be chosen without loops; loops remain available in the ground set for quota completion. Define

$$\phi(C) = |C \cap G_0| + \sum_{i=1}^{s} \max\{0, |C \cap S_i| - \beta\}.$$

Completing the quotas has cost $\beta s + \phi(C)$, so minimizing ϕ over resilient cores of size at most q yields an optimum. The completion cost must be preserved: in $U_{1,4}$ with $k = 1$ and $S_1 = \{c, d\}$, the core $\{a, b\}$ completes to four elements, while $\{c, d\}$ is already optimal.

Color-capped core. Set $H = \binom{r+k-1}{k}$ and

$$c_h = (h-1)H + q, \qquad \theta_h = (q+1)c_h.$$

Lemma 5.5, specialized to unit weights and all elements vulnerable, supplies a resilience-preserving same-cardinality exchange from any h-uniform candidate set of size at least c_h. Start with the nonloops $E^\circ = \{e : \operatorname{rank}_M(\{e\}) = 1\}$, regarding $G_0, S_1, \dots, S_s$ as colors. On a rank-h domain A, scan its elements, accepting them into an h-uniform set Z with at most c_h elements per color. Stop at $|Z| = \theta_h$ or when the scan is exhausted. If $h = 1$, return Z; otherwise return Z together with the recursively computed cores of

$$A \cap \mathrm{cl}_M(Q), \qquad Q \in \tbinom{Z}{h-1}.$$

Before h elements have been accepted, the color cap cannot bind because $c_h \geq q \geq r \geq h$. Thus $\operatorname{rank}(Z) = h$ and every child has rank $h-1$.

Optimum preservation. We induct on h to prove that every resilient B with $|B| \leq q$ can be replaced by a resilient set of the same size, unchanged outside A, with A-part in the returned core and nonincreasing ϕ. Write $U = Z$ when $h = 1$ and otherwise let U be the union of the child domains. As in Lemma 5.7, dependence-rejected elements lie in U, and $Z \subseteq U$.

Consider $x \in (B \cap A) \setminus U$. If x was examined by the scan, it was rejected at its color cap. The c_h retained elements of that color admit a replacement by Lemma 5.5, preserving all color counts and

hence ϕ. Otherwise the total threshold was reached. With $Y = B \setminus \{x\}$, discard the color G_0 and any group satisfying $|Y \cap S_i| \geq \beta$. At most q colors are discarded, since $|Y| \leq q - 1$ and $\beta \geq 1$; they cover at most qc_h points of Z. At least c_h candidates remain in unsaturated groups, so Lemma 5.5 supplies an exchange that does not increase ϕ; removing x cannot increase it either.

Repeated exchanges place the selected elements in U, finishing the rank-one case. At higher rank, apply the induction hypothesis to the child domains. The decreasing residual-set argument from the proof of Lemma 5.7 handles overlapping children, and each exchange preserves the cost inequality. This proves the invariant.

Running time. The size of the resulting core is bounded by L_r where

$$L_1 = \theta_1, \qquad L_h = \theta_h + \binom{\theta_h}{h-1} L_{h-1} \quad (2 \leq h \leq r).$$

This is a function of r, k; the recursion depth is at most r, and each node performs polynomially many oracle operations for fixed r, k. Enumerating all core subsets of size at most q, checking up to 2^q failures for each, minimizing ϕ, and completing the quotas gives the stated running time and an optimum placement. □

Corollary 6.3. *For arbitrary directed state graphs, minimum unit-cost dedicated-actuator placement (at most one actuator per state) that tolerates any failure of at most k actuators is fixed-parameter tractable in the maximum-matching deficiency d plus k.*

Proof. Use the dual of the state matching transversal matroid, with rank $r = d$, the state maximum-matching deficiency. Surviving matching feasibility is spanning in this dual, and surviving accessibility is equivalent to selecting at least $k + 1$ states in each source SCC. The latter sets are disjoint, so Theorem 6.2 applies. □

7 Conclusion and open problems

The one-failure unit-cost problem admits a cubic-time 3/2-approximation on arbitrary state digraphs, while two independent core constructions give exact fixed-parameter algorithms: one for weighted resilient spanning with safe and vulnerable elements, and the other for unit-cost resilient spanning with an arbitrary number of disjoint source quotas. Proposition 3.1 transfers the matroid results to dedicated-actuator placement under the specified failure semantics.

The weighted result permits mixed failure statuses but does not incorporate arbitrary source quotas; the quota result permits any number of groups but assumes unit costs and failures of all selected elements. Combining nonnegative weights, mixed failure statuses, and source quotas is open. The optimal polynomial-time approximation guarantee for general state digraphs also remains unresolved. Example 4.6 demonstrates equality for one relaxation choice, without implying a hardness-of-approximation threshold.

Code availability. Source code and verification scripts are publicly available at `https://github.com/dongxuelian2/robust-structural-controllability`. Reproducible versions of the approximation implementation and the parameterized implementation are identified by the linked Git commit revisions. Licensing terms, if any, are those stated in the repository; no archival DOI is claimed.

Data availability. No external empirical dataset is required for the mathematical results. Computational verification inputs are generated or documented by the publicly available scripts and source files cited above.

AI assistance and author responsibility. OpenAI-based language-model tools were used for editorial assistance, argument exploration and checking, and the preparation of verification code. The authors retain responsibility for the mathematical statements, the reproducibility of the code, and the final manuscript.

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